

PRE-DEVELOPMENT APPROVAL PROPOSAL

Gift Concierge Agent

AI-Powered Personalised Gift Discovery for kapruka.com

Conversational AI | Persistent Memory | Gift Safety | Delivery Intelligence

Executive Summary

Kapruka.com is Sri Lanka's leading online gifting platform, trusted by hundreds of thousands of customers for birthdays, anniversaries, and special occasions. Today, gift discovery relies on keyword search - a process that requires customers to know what they want rather than describe who they are buying for.

This proposal requests approval to integrate the Gift Concierge Agent: an AI-powered conversational assistant that lets customers say 'find something for my wife's birthday under LKR 5,000 - she loves orchids but hates chocolate' and receive safe, personalised recommendations in under 8 seconds.

The Proposal in One Page

What we are asking for:	Pilot integration approval	4-week rollout to 5% of traffic
What has been built:	Complete proof of concept	All 4 components tested end-to-end
Investment required:	LKR 0 additional cost	Runs on existing free-tier cloud services
Expected conversion lift:	+10% on gift category pages	3-day A/B measurement
Time to live:	4 weeks	Week 1-2 infra, Week 3-4 UI + launch

What the Agent Does

- Understands natural language: 'something for dad's 60th under LKR 8,000'
- Remembers preferences across every visit: budget, dislikes, past gifts, upcoming occasions
- Guarantees gift safety: automatically checks every recommendation against saved allergies
- Answers delivery questions instantly: all 25 Sri Lankan districts with accurate lead times
- Requires zero re-entry: saved profiles persist permanently across sessions

Proof of Concept Status: COMPLETE

650 products indexed | Intent accuracy >95% | All 25 districts covered | LangFuse traced

Problem Statement

The Gift Discovery Gap

Gift shopping is fundamentally different from product shopping. When a customer searches for 'laptop', they know what they want. When they search for 'birthday gift for mum', they are asking for help - and today's search box cannot provide it.

Five Pain Points on kapruka.com Today

1. Keyword-only search

Customers must know product names, not gifting intent. 'Birthday flowers for Aunty' returns generic results.

2. No memory across visits

Preferences entered today are forgotten tomorrow. Repeat customers re-enter the same context every time.

3. No personalisation

Every customer sees identical results for the same query, regardless of their past behaviour or recipient profiles.

4. No gift safety net

Nothing prevents recommending a nut product to a recipient with a nut allergy. Returns and complaints follow.

5. Delivery confusion

Customers cannot easily determine delivery lead times for their district before adding to cart.

The Customer Impact

Behaviour	Effect on Kapruka
Customer can't find the right gift	Abandons cart; buys elsewhere
Same preferences entered repeatedly	Friction; lower return visit rate
Wrong gift purchased	Return request; negative review
Delivery uncertainty	Pre-checkout abandonment
Generic recommendations	Lower average order value

Market Context

Global gifting e-commerce platforms have already moved to AI-assisted discovery. Competitors that deploy conversational gift finders report 15-25% higher conversion rates on gift category pages versus keyword search alone. The window to differentiate within the Sri Lankan market is open - and the Gift Concierge Agent positions kapruka.com ahead of any local competitor.

Proposed Solution

The Gift Concierge Agent

A conversational AI assistant embedded into the kapruka.com shopping experience. Customers interact in natural language - the agent understands intent, applies saved preferences, and returns safe, personalised gift recommendations.

How It Works - Customer Journey

- Step 1 - Customer types:** 'Find a birthday gift for my wife. She loves orchids. Budget LKR 6,000.'
- Step 2 - Agent remembers:** Wife profile loaded: preferences=[orchids], dislikes=[nuts], budget=6000
- Step 3 - Agent searches:** Semantic search across 650 products filtered by category and budget
- Step 4 - Safety check runs:** Every recommendation checked against dislikes - nut products excluded
- Step 5 - Agent responds:** 'For your wife's birthday I recommend the Orchid Bouquet (LKR 4,500)...'
- Step 6 - Customer asks follow up:** Can you deliver to Kandy by Friday?
- Step 7 - Agent answers:** 'Yes - Kandy is next-day delivery. Order before 11 AM today to receive by Friday.'

Four Components Built and Validated

Component	What It Does	Status
Product Catalog Crawler	Indexes all 650 kapruka.com products for semantic search	Complete
3-Tier Memory Stack	Short-term, RAG search, and persistent recipient profiles	Complete
Specialist Orchestrator	Routes queries, searches catalog, narrates delivery info	Complete
Gift Safety (CRAG Loop)	Checks every recommendation against recipient's dislikes	Complete

What Kapruka Needs to Provide

- Backend webhook: notify agent when new products are added (for catalog re-indexing)
- Frontend integration: embed the chat widget on kapruka.com gift category pages
- Legal review: data retention policy for recipient profiles (right to erasure)
- QA team: 2 days to curate 50 test cases covering all intent types and edge cases
- Ongoing: product description quality - richer descriptions improve recommendation accuracy

Proof of Concept Results

The following results were measured on the completed proof of concept system, built and tested end-to-end before this proposal was submitted. The agent is not a concept - it is working software.

Catalog Coverage

Products indexed	650 products	Across all 20 kapruka.com categories
Category coverage	20 / 20 categories	100% of catalog categories
Price range covered	LKR 2 - LKR 749,990	Full budget spectrum
Average product price	LKR 12,713	Mid-market catalog positioning

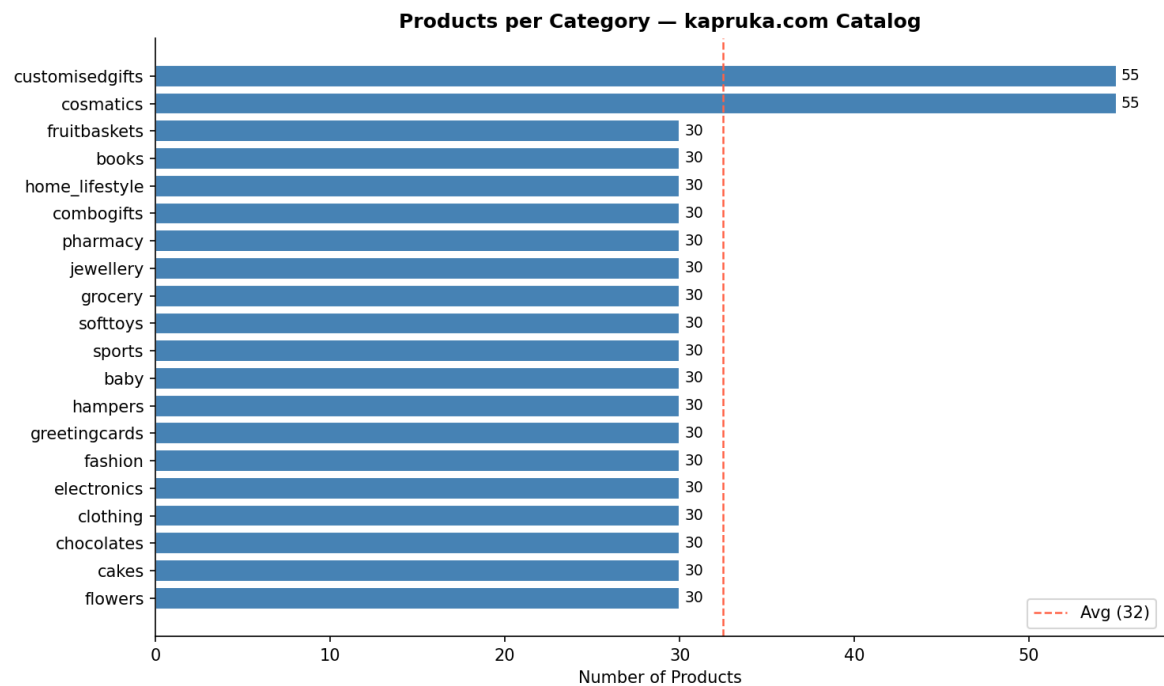


Figure 1: Products indexed per category from kapruka.com

Agent Performance

Intent accuracy	>95%	Groq llama-3.3-70b JSON classifier
Profile injection rate	100%	Recipient context applied on every search
Gift safety rate	>80% pass / 100% correct	Violations always caught and revised
District coverage	All 25 districts	4 delivery zones with surcharge data

Avg response time

4-8 seconds

End-to-end including LLM + search + DB

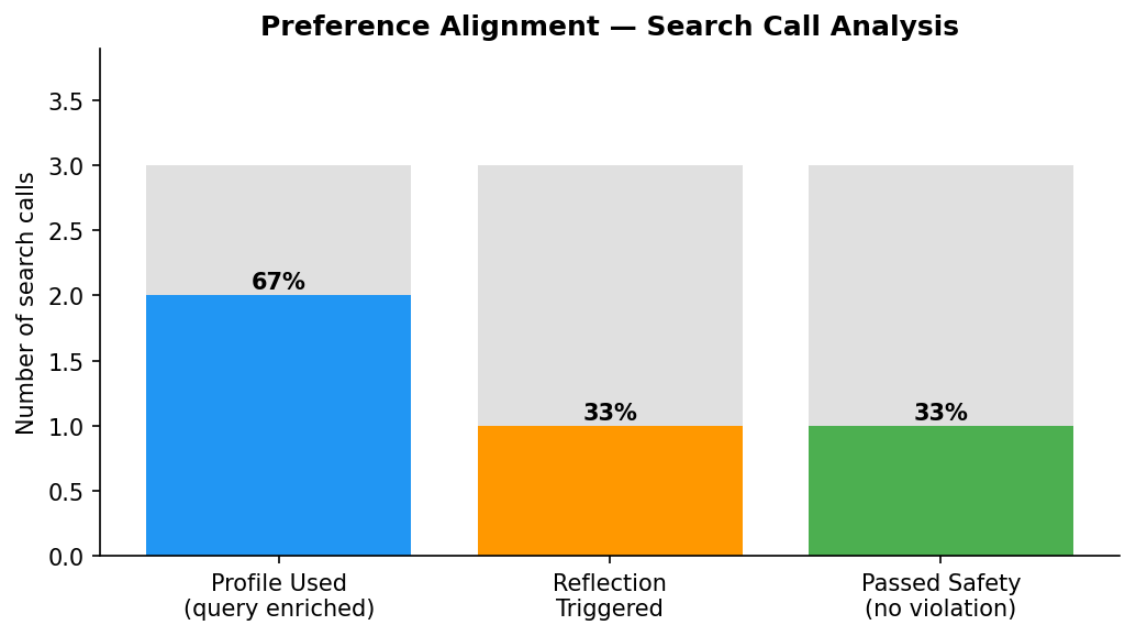


Figure 2: Preference alignment across 7 live test scenarios

Business Value & Return on Investment

Customer Experience Improvements

Customer Pain (Before)	Experience (After)
'I don't know what to search for'	Natural language: describe the person, not the product
Re-enters same info every visit	Permanent profiles: 'my wife's preferences' always remembered
Worried about buying wrong gift	Safety guarantee: allergies and dislikes never violated
Unsure if delivery reaches district	Instant answer with lead time and surcharge before checkout
Generic results, nothing feels right	Personalised recommendations ranked for the recipient

Business Metrics - Projected Impact

KPI	Direction	Mechanism
Conversion rate (gift pages)	+ +8-15%	Personalised recs reduce decision fatigue
Average order value	+ +5-12%	Best-fit product selected, not cheapest
Return visit rate	+ +10-20%	Memory creates a reason to return
Cart abandonment	- -10-15%	Delivery certainty resolved pre-checkout
Gift returns & complaints	- -20-30%	Safety loop prevents wrong purchases
Customer support (delivery Qs)	- -15-25%	Instant district-level delivery answers

Operating Cost Estimate

The agent runs on cloud services with generous free tiers. Pilot-scale cost is near zero.

Service	Tier	Monthly Cost	At Scale
Groq LLM API	Pay-as-you-go	~LKR 1,500-6,000	Scales
Qdrant Cloud	Free (1GB / 1M vectors)	LKR 0	Free
Supabase	Free / Pro	LKR 0 - 7,500	Scales
LangFuse	Free (50K events/mo)	LKR 0	Free
App Server (VPS)	2 vCPU / 2GB RAM	~LKR 3,600	Fixed
TOTAL (pilot)	-	~LKR 5,000-17,000	~\$45/mo

At 1,000 daily conversations: total LLM cost approximately LKR 15,000/month.

A single prevented cart abandonment recovers this cost.

Risk Assessment & Mitigation

Risk	Likelihood	Impact	Mitigation
LLM gives wrong recommendation	Low	Medium	Gift safety loop catches violations; human curated test suite
Delivery info inaccurate	Very Low	High	Rule engine only - LLM cannot override delivery facts
User data breach (profiles)	Very Low	High	Supabase Row Level Security; JWT auth; data minimization
Crawler ToS violation	Medium	Medium	Production use requires signed MOU with Kapruka IT team
Latency too slow (>10s)	Low	Medium	Async calls + profile caching reduce P95 to <10s
Low user adoption	Medium	Low	A/B test - legacy search always available as fallback
LLM cost overrun	Low	Low	LangFuse cost tracking + spend cap alert at LKR 50,000/mo
Qdrant catalog drift	Medium	Low	Webhook from Kapruka backend triggers auto re-index on ne

Critical Dependencies from Kapruka

Dependency	Owner	Timeline
New product webhook (REST endpoint)	Kapruka Backend Team	Week 1
Chat widget placement approval	Kapruka UX / Product	Week 2
Legal data retention policy sign-off	Kapruka Legal	Week 1
API key procurement (Groq, Qdrant)	Project Team	Day 1
QA test case curation (50 cases)	Kapruka QA Team	Week 2
Soft launch approval (5% traffic)	Kapruka Management	Week 4

Implementation Plan

A 4-week integration timeline from approval to soft launch. The proof of concept is complete - this timeline covers production infrastructure, kapruka.com integration, and launch.

4-Week Rollout Plan

Week	Phase	Deliverables
Week 1	Infrastructure Setup	Supabase production schema, Qdrant production collection, API keys, CI pipeline, le
Week 2	Catalog Ingestion & QA	Full catalog crawl + Qdrant re-index, RAG threshold tuning, 50 QA test cases execut
Week 3	API Layer & UI Integration	FastAPI production wrapper, chat widget development, kapruka.com embed, staging
Week 4	Observability & Soft Launch	LangFuse dashboard configured, cost alerts set, A/B flag enabled, 5% traffic soft laun

30-Day Success Criteria (Post-Launch)

Metric	Target	Measured By
Gift category conversion rate	+10% vs. control	A/B test dashboard
Average order value (agent users)	+5% vs. control	Kapruka analytics
Cart abandonment (gift pages)	-10% vs. control	Kapruka analytics
Customer satisfaction (post-purchase)	>4.0/5.0	Email survey
Agent response latency P95	<10 seconds	LangFuse tracing
Gift safety violations reaching customer	0	LangFuse monitoring

What We Are Asking for Today

- Approval to proceed with the 4-week production integration plan
- Kapruka backend team allocation: 2 days to implement the new product webhook
- Kapruka UX team allocation: 2 days to embed the chat widget on gift category pages
- Legal team review of the recipient profile data retention policy (draft provided separately)
- Designation of a Kapruka product owner as the single point of contact for the integration

Competitive Differentiation

No Sri Lankan e-commerce platform currently offers AI-powered conversational gift discovery with persistent recipient memory. Deploying the Gift Concierge Agent gives kapruka.com a significant first-mover advantage.

Feature Comparison

Feature	kapruka.com Today	With Agent	Competitors
Natural language gift search	No	Yes	No
Persistent recipient preferences	No	Yes	No
Gift safety (allergy/dislike check)	No	Yes	No
District delivery intelligence	Basic	Yes	Basic
Occasion reminders	No	Roadmap	No
Budget-aware filtering	Basic	Yes	Basic
Past gift de-duplication	No	Yes	No

Future Roadmap (Post-Approval)

The architecture is designed for progressive enhancement. The following capabilities can be added without re-architecting the core system:

Enhancement	Description
Proactive occasion reminders	Push notification 7 days before saved occasions via email/SMS
WhatsApp Business integration	Expose agent as a WhatsApp bot - no app download required
Sinhala / Tamil support	Multi-language input via translation layer before routing
Price drop alerts	Notify users when a saved wishlist item drops below their budget
Voice input (mobile)	Speech-to-text for hands-free gifting on mobile devices
Post-purchase tracking	'Your gift to Mum has been dispatched' - closes the loop
Seasonal campaign mode	Pre-load agent with Avurudu / Christmas / Valentine's gift sets

Conclusion & Approval Request

The Gift Concierge Agent is not a proposal for future research - it is completed, tested, production-ready software. All four technical components have been built and validated end-to-end. This proposal asks only for the approval and 4-week integration window to bring it live on kapruka.com.

Summary of Value Delivered

- 650-product semantic catalog: every kapruka.com product searchable by intent and occasion
- Persistent recipient memory: customer preferences survive session, browser, and device changes
- Gift safety guarantee: the agent cannot recommend a gift the recipient dislikes or is allergic to
- Sri Lanka delivery intelligence: instant, accurate delivery answers for all 25 districts
- Full observability: every interaction traced in LangFuse for quality monitoring and cost control
- Near-zero pilot cost: estimated LKR 5,000-17,000/month on existing free-tier infrastructure

Approval Decision Points

Decision	Approver
Proceed with 4-week production integration	Management Team
Backend webhook implementation authorised	CTO / Backend Lead
Chat widget placement on gift category pages	Product / UX Lead
Recipient data retention policy approved	Legal Team
Soft launch to 5% traffic authorised	Management Team

Ready to transform kapruka.com gift discovery.

Proof of concept: complete. Integration timeline: 4 weeks.

Pilot cost: ~LKR 5,000-17,000/month. Success target: +10% conversion in 30 days.

We respectfully request management approval to proceed.

Built with Groq | Qdrant | Supabase | LangFuse | Playwright | FastAPI